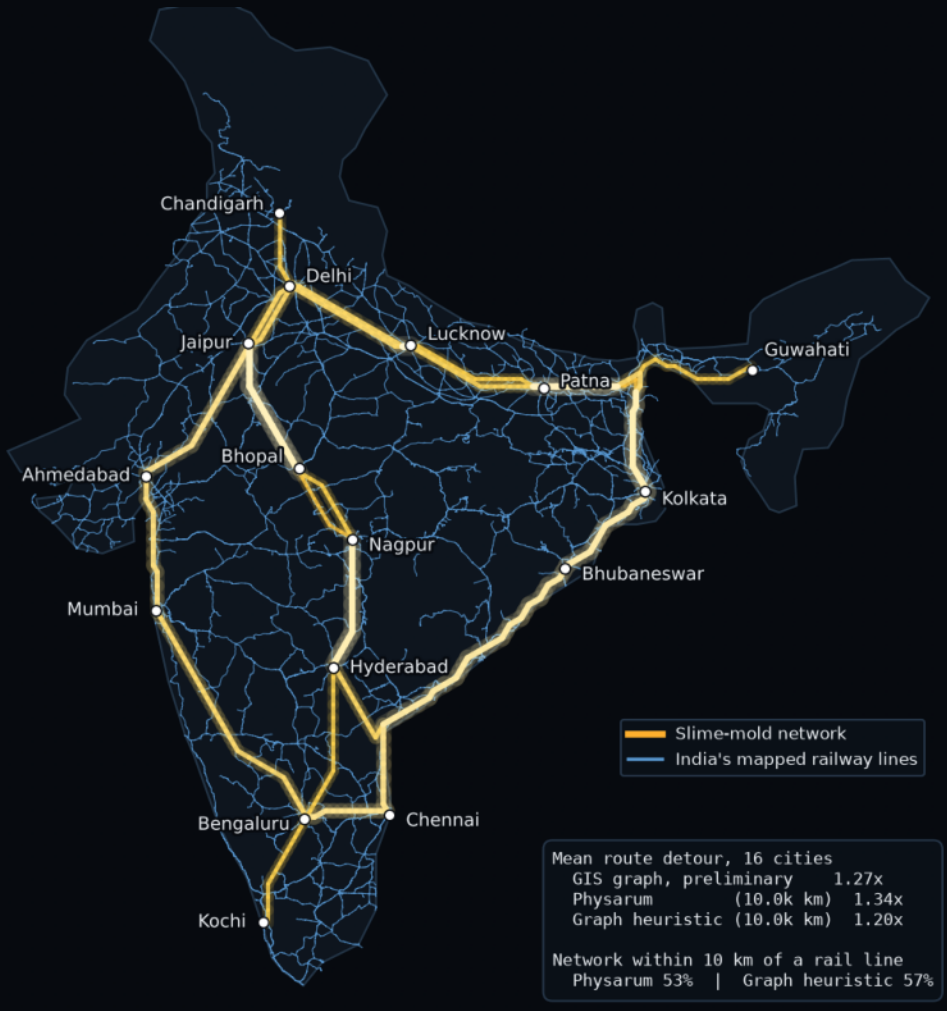
I asked a slime-mold algorithm to connect 16 Indian cities.

Then I tested it against India's real railway map, and against simple graph algorithms.



Swipe →

FLOW-DRIVEN ADAPTATION

How the network grows

1

Push flow

Random city pairs send current through the whole mesh of ~35,000 candidate routes (a global flow solve).

2

Reinforce

Each route thickens or thins based on its own flux.

3

Fade

Unused routes decay. What survives is the network.

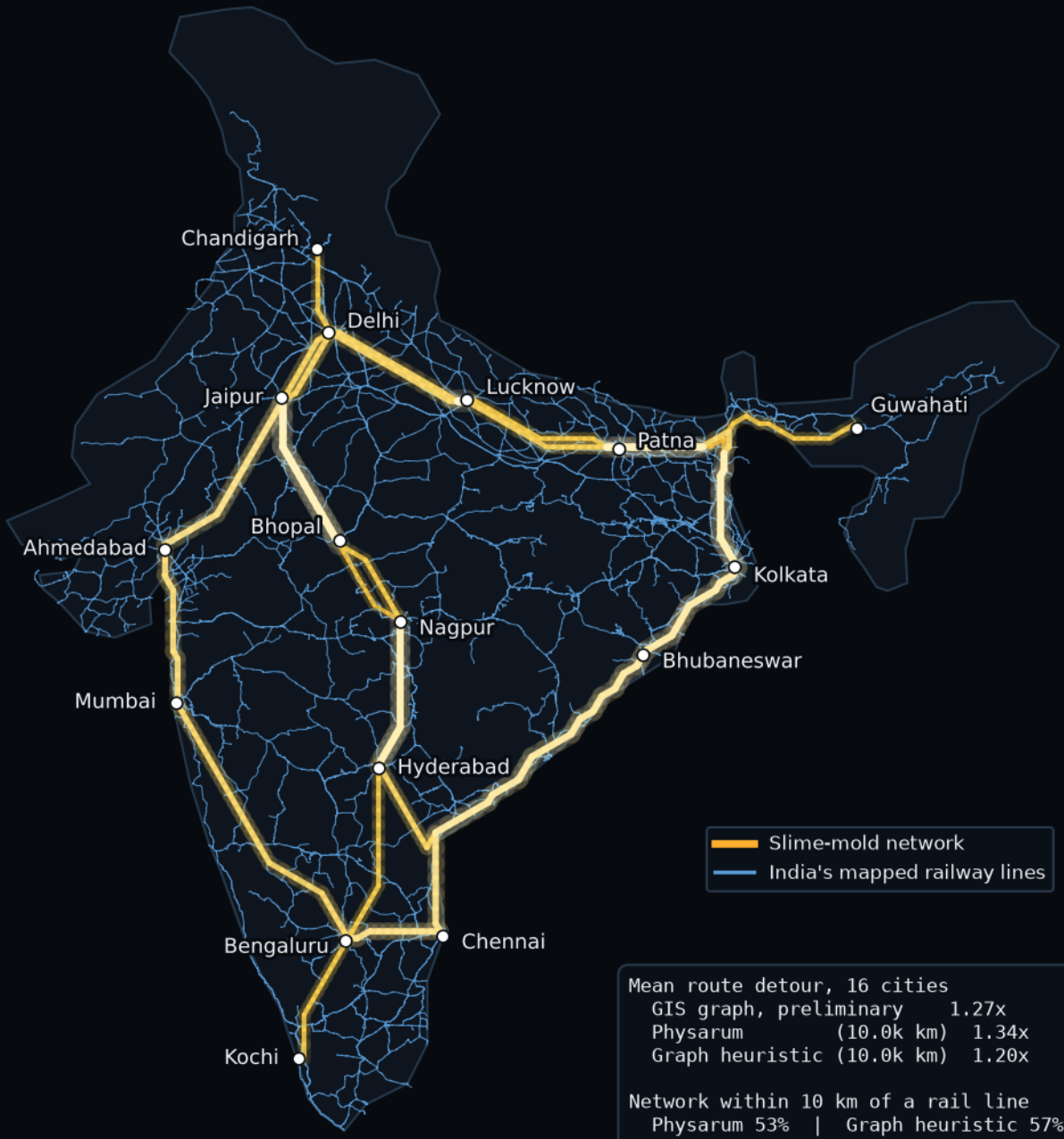
$$D \leftarrow D + \Delta t\,(f(|Q|) - D), \quad f(Q) = \frac{Q^{\gamma}}{1 + Q^{\gamma}}$$

D = tube thickness, Q = flow, γ = 1.6. Model: Tero et al., Science 2010 (Tokyo rail).

Slime mold vs. India

Can Physarum-inspired flow rules connect 16 Indian cities?

Step 2 | Compare with India's mapped railway lines




```
Mean route detour, 16 cities
  GIS graph, preliminary    1.27x
  Physarum        (10.0k km)  1.34x
  Graph heuristic (10.0k km)  1.20x

Network within 10 km of a rail line
  Physarum 53%  |  Graph heuristic 57%
```

Model: Tero et al., Science 327 (2010) | Python, NumPy, SciPy

Rail lines: datta07/INDIAN-SHAPEFILES (third-party GIS). Outline and terrain costs are approximate.

ONE MORE TEST

Is slime mold the best network design?

MEAN ROUTE DETOUR BETWEEN THE 16 CITIES (LOWER IS BETTER)

1.34×	1.20×	1.27×
Physarum	**Graph heuristic**	**Stitched GIS graph**
10,035 km network	10,016 km network	entire network*

A simple graph heuristic found shorter routes at similar total length.

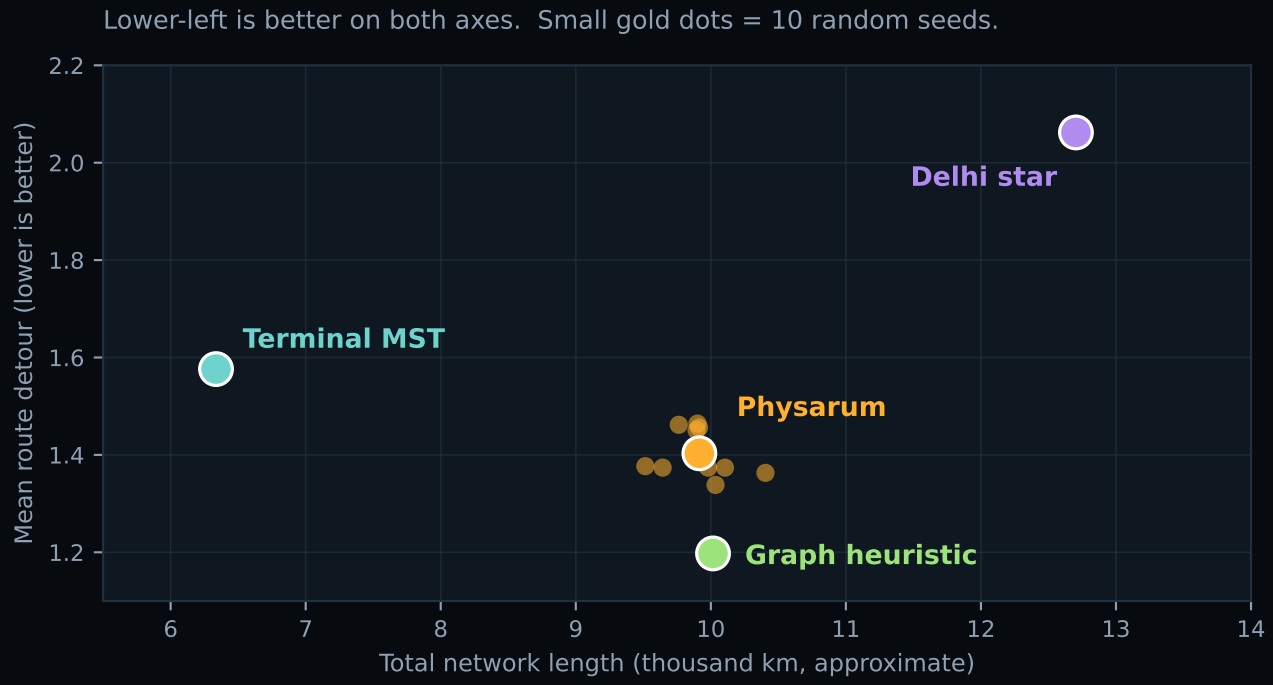
Static, city-excluding zones (10-seed mean). Across 300 150 km blocked zones, Physarum kept all 120 city pairs connected 96% of the time, vs 93% for the graph heuristic and 80% for a plain spanning tree.
Similar results are not a proven resilience advantage.

Takeaway: reinforcing busy routes grows connected, redundant networks. It does not magically beat engineers.

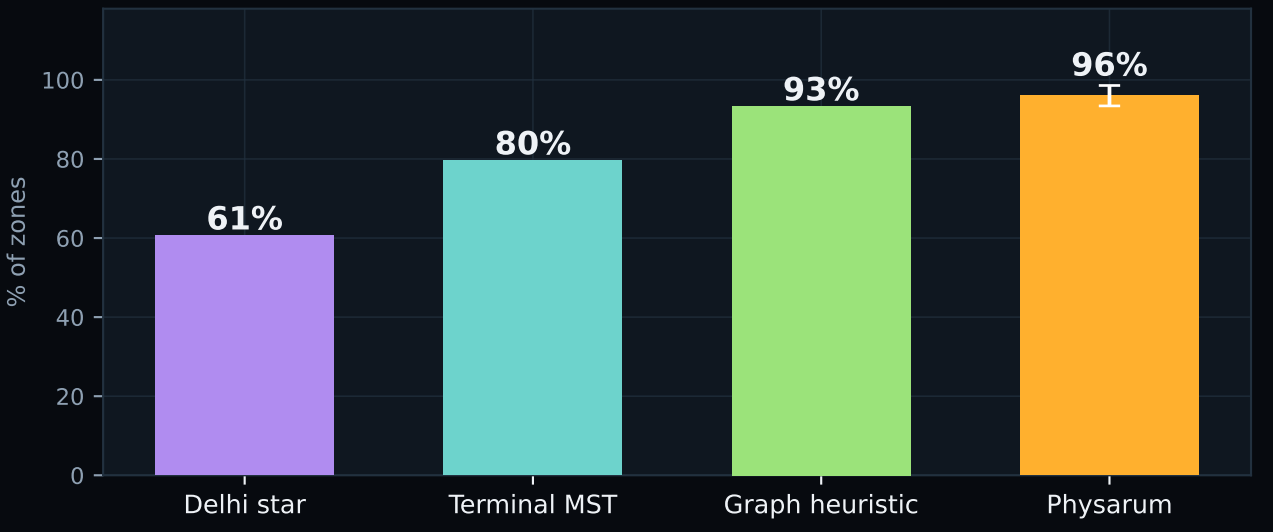
*GIS graph = the whole mapped network (about 86,000 km of line, parallel tracks included), Not an equal-cost comparison; junctions unverified. Seed-7 model, approximate terrain.
Rail data: datta07/INDIAN-SHAPEFILES (third-party GIS). Model: Tero et al., Science 2010.

TRADE-OFFS, NOT A WINNER

Shorter routes or more resilience?



All 120 pairs connected after 150-km static damage



Static, city-excluding lattice zones; no re-adaptation. Physarum bar: mean of 10 seeds, error bar = 1 SD. Approximate geography and synthetic terrain.

SLIME MOLD vs. GRAPH ALGORITHMS

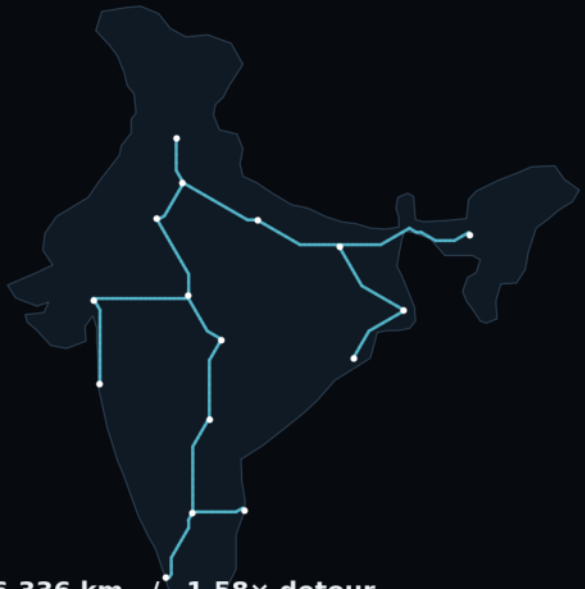
16 Indian cities | Same candidate grid | Seed 7

PHYSARUM



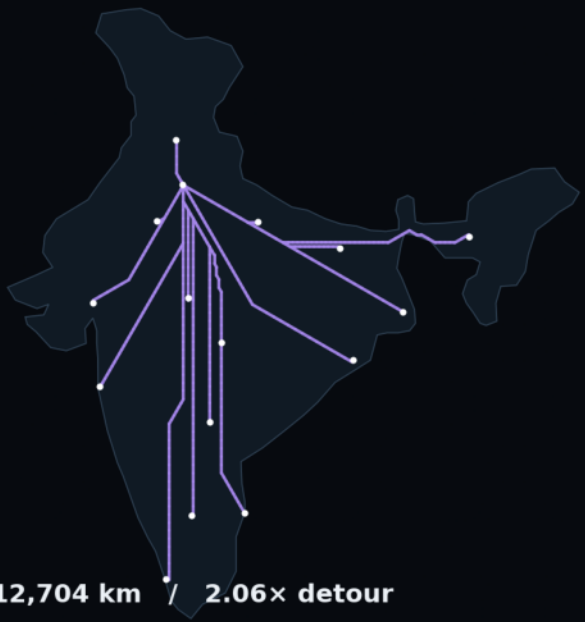
10,035 km / 1.34× detour

MST



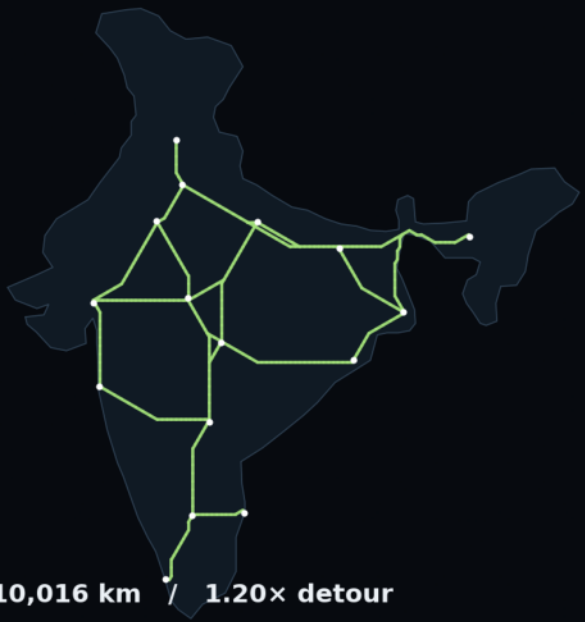
6,336 km / 1.58× detour

DELHI SHORTEST-PATH UNION



12,704 km / 2.06× detour

AUGMENTED MST



10,016 km / 1.20× detour

FINDING: THE PHYSARUM MODEL DOES NOT DOMINATE

The augmented MST achieves shorter routes at similar modeled length.

Illustrative geography and synthetic terrain, NOT actual railway tracks.

HONEST SCORECARD

What this experiment shows

SHOWS

- Flux-driven reinforcement grows a connected, redundant 16-city network (10 of 10 seeds)
- Under random damage it is about as resilient as a purpose-built graph heuristic, and more resilient than these tree baselines

DOES NOT SHOW

- That slime mold beats engineers: a simple graph heuristic found shorter routes
- That it reproduces Indian Railways: no verified junctions, topology or demand
- Real elevation, rivers or construction cost

NEXT

- Real population and freight demand
- Rivers and elevation in the cost model
- Validate physical junctions and station-level routes

Public v4.1 code; extra results in the local v4.2 supplement.

Model: Tero et al., Science 327 (2010). Rail GIS: datta07/INDIAN-SHAPEFILES (third-party).